CH-II-Exemples-1-2-P18 1

Context

Produced by a self-taught mathematic amateur.
I document my solutions to exercises and my explorations here.
Corrections by *DeepSeek*.
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Chapter II: The theory of convergence

Bounding the general term

Show that if $\theta \in]0, 2\pi[$, then

$$\left|\sum_{n=1}^{p} \sin(n\theta)\right| < \csc\left(\frac{\theta}{2}\right)$$

Indeed,

$$\sin(n\theta)\sin\left(\frac{\theta}{2}\right) = \frac{1}{2}\left[\cos\left(\frac{2n-1}{2}.\theta\right) - \cos\left(\frac{2n+1}{2}.\theta\right)\right]$$

$$\sum_{n=1}^{p} \sin(n\theta)\sin\left(\frac{\theta}{2}\right) = \frac{1}{2}\sum_{n=1}^{p}\left[\cos\left(\frac{2n-1}{2}.\theta\right) - \cos\left(\frac{2n+1}{2}.\theta\right)\right] \Rightarrow$$

$$\sum_{n=1}^{p} \sin(n\theta)\sin\left(\frac{\theta}{2}\right) = \frac{1}{2}\left[\cos\frac{\theta}{2} - \cos\left(\frac{2p+1}{2}.\theta\right)\right] = \sin\left(\frac{p\theta}{2}\right)\sin\left(\frac{p+1}{2}\theta\right) \;;\; \cos(a) - \cos(b) = 2s$$

$$\left|\sum_{n=1}^{p} \sin(n\theta)\sin\left(\frac{\theta}{2}\right)\right| = \left|\sin\left(\frac{p\theta}{2}\right)\sin\left(\frac{p+1}{2}\theta\right)\right| < 1, \forall\theta \in]0, 2\pi[\Rightarrow$$

$$\left|\sin\left(\frac{\theta}{2}\right)\sum_{n=1}^{p}\sin(n\theta)\right| < 1 \Rightarrow \left|\sum_{n=1}^{p}\sin(n\theta)\right| < \csc\left(\frac{\theta}{2}\right)$$

Series study

The real sequence f_n is monotonically decreasing and $f_n \to 0$.

The series $\sum f_n \sin(n\theta)$

We infer that if f_n is monotonically decreasing and $f_n \to 0$, then, by **Dirichlet's test**, $\sum f_n \sin(n\theta)$ is (ordinarily) convergent for every $\theta \in \mathbb{R}$.

Error

Converges for every $\theta \in \mathbb{R}$ except $\theta = \frac{\pi}{2}$.

Correction (AI)

For $\theta = \frac{\pi}{2}$,

$$\sum_{n=1}^{\infty} f_n \sin(n\pi/2) = \sum_{n=0}^{\infty} f_{2n+1} \sin\left((2n+1).\frac{\pi}{2}\right) = \sum_{n=0}^{\infty} (-1)^n . f_{2n+1}$$

Even though we know nothing a priori about the convergence of the extracted series $\sum f_{2n+1}$, since $\lim_{n\to\infty} f_{2n+1} = 0$ (it is still monotonically decreasing), the convergence of the series $\sum (-1)^n f_{2n+1}$ is guaranteed by **Leibniz's test**, which is just a special case of Dirichlet's test in which the general term of the bounded sequence is simply $(-1)^n$.
The same applies for every $\theta = (2m+1)\pi/2$.
-If $n = 2k$: $\sin\left((2m+1)\frac{n\pi}{2}\right) = \sin\left((2m+1)k\pi\right) = 0$
-If $n = 2k+1$: $\sin\left(\frac{(2k+1)(2m+1)\pi}{2}\right) = (-1)^{m+k}$.
The extracted subsequence series becomes:

$$\sum_{n=1}^{\infty} f_n \sin\left(n\frac{(2m+1)\pi}{2}\right) = \sum_{k=0}^{\infty} (-1)^{k+m} . f_{2k+1} = (-1)^m \sum_{k=0}^{\infty} (-1)^k . f_{2k+1}$$

The same reasoning applies.

Thus, the series $\sum f_n \sin(n\theta)$ converges for every $\theta \in \mathbb{R}$.

The series $\sum f_n \cos(n\theta)$

We also infer that since f_n is monotonically decreasing and $f_n \to 0$, then, again by **Dirichlet's test**, $\sum f_n \cos(n\theta)$ is (ordinarily) convergent for every $\theta \in \mathbb{R} \setminus \{2\pi k, k \in \mathbb{Z}\}$.

Remark

For the **cosine** case, it is different: when $\theta = 2\pi$, we get $\sum f_n \cos(2n\pi) = \sum f_n$. However, we know nothing about the convergence of this series. The same applies for every other even multiple of π, since $\cos(2nk\pi) = 1$.

Example 2: the alternative versions

The series $\sum (-1)^n f_n \cos(n\theta)$

Substituting $\theta+\pi$ for θ,

$$\sum f_n \cos(n(\theta+\pi)) = \sum (-1)^n f_n \cos(n\theta)$$

The series $\sum f_n \cos(n(\theta+\pi))$ converges for every $\theta+\pi \in \mathbb{R} \setminus \{2\pi k, k \in \mathbb{Z}\} \Leftrightarrow \theta \in \mathbb{R} \setminus \{(2k-1)\pi, k \in \mathbb{Z}\}$.

The series $\sum (-1)^n f_n \sin(n\theta)$

Substituting $\theta+\pi$ for θ,

$$\sum f_n \sin(n(\theta+\pi)) = \sum (-1)^n f_n \sin(n\theta)$$

The series $\sum f_n \sin(n(\theta+\pi))$ converges for every $\theta \in \mathbb{R}$. Consequently, the alternative version also converges for every real θ.